

Pattern Explanation:

The Flyweight Pattern is a structural design pattern that helps reduce memory usage by sharing common objects instead of creating a new object every time it is needed. In the emoji chat system, emojis like 😂, ❤️, and 😭 are used repeatedly by many users. Instead of creating separate objects for the same emoji, the system stores one shared object for each emoji and reuses it whenever that emoji is sent in the chat. This makes object creation more efficient and avoids unnecessary duplication.

Justification:

We used the Flyweight Pattern because chat applications usually contain many repeated emojis. If the system created a new object every time a user sent the same emoji, it would waste memory and reduce performance, especially in large chats with many users. By sharing the same emoji object, the system saves memory, improves performance, and makes resource management more efficient while keeping the chat functionality the same.